



UC-7427

M. C. A. (Integrated) (Sem. II)
Examination, June - 2023

IMCA-202

COMPUTER ARCHITECTURE

Time allowed : Three Hours

Maximum Marks : 70

Min. Pass Marks : 28

Note : This question paper contains three sections as under :

Section-A**Max. Marks - 10**

This section shall have 01 compulsory question comprising 10 questions (maximum 20 words answer) taking two questions from each unit. Each question shall be of one mark and total marks of this section will be 10. This section will be compulsory in the paper.

Section-B**Max. Marks-25**

This section will carry 25 marks with equally divided into five long answer type questions (answer about in 250 words) and students are instructed to attempt five questions by selecting one question from each unit.

Section-C**Max. Marks-35**

Will carry 35 marks with five long answer type questions comprising one compulsory question (question no. 12) of 15 marks and four questions of 10 marks each. Students are instructed to attempt total three questions with one compulsory question (answer about in 500 words) and any two more questions (answer about in 400 words) out of remaining four questions.

UC-7427]

2

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SECTION - A

- 1 Answer in one line
- (i) RISC
 - (ii) CMOS
 - (iii) Fetch cycle
 - (iv) ALU
 - (v) Virtual Memory
 - (vi) CPU
 - (vii) DTL
 - (viii) MOSFET
 - (ix) DMA
 - (x) External Devices

**SECTION - B
UNIT - I**

- 2 Explain Instruction cycle in detail

OR

- 3 Explain Bus systems of basic computer, how they are inter connected.

UNIT - II

- 4 Explain :
- (a) Addressing modes
 - (b) Instruction formats

OR

- 5 Explain CISC Architecture.

UC-7427]

3

[Contd...

UNIT - III

- 6 Explain virtual memory organization.

OR

- 7 Explain cache memory. Describe Hit Ratio.

UNIT - IV

- 8 Explain Programmed I/O.

OR

- 9 Describe : (i) DMA (ii) I/O Channels

UNIT - V

- 10 Describe : (i) MOSFET (ii) TTL

OR

- 11 Describe : (i) RTL (ii) DTL

SECTION - C

- 12 Explain Arithmetic and Logical Micro-operation.

- 13 Describe Interrupt. How it is useful?

- 14 What is Interrupt driven I/O.

- 15 Describe parallel architecture.

- 16 Describe MOSFET and CMOS. Also write difference between them.